PHYS 365/369 — Midterm Fall 2023 — Worked Solutions

20 questions. Every step uses the formula sheet printed at the back of the F2023 midterm. Anything that is *not* on that sheet is put in an orange **REMEMBER** box. Every final answer was checked against the official answer key. (The exam's own numbering skips "Q10": the scenario box says "Questions 11, 12 and 13", but they are numbered 10, 11, 12. The exam's numbering is used here.)

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	6	C	11	A	16	A
2	B	7	C	12	A	17	D*
3	A	8	D	13	B	18	B
4	D	9	D	14	B	19	C
5	B	10	C	15	B	20	D

NOTE ON THE ANSWER KEY

*Q17: the correct value works out to $\dfrac{\mu_2}{\mu_1} = \dfrac{1}{81}$. The key marks D, which is printed as $\frac{1}{8}$. That is almost certainly a typo for $\frac{1}{81}$. See the full solution.

REMEMBER — not on the formula sheet

- Adding two waves with **different** amplitudes: use phasors. Add the components, or use $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\Phi}$ (Q1). The sheet's $2y_m\cos\left(\frac{\Phi}{2}\right)$ only works for **equal** amplitudes.
- SHM: $a = -\omega^2 x$. Acceleration is always opposite to displacement (Q8).
- Waves carry energy, not matter (Q6).
- Pulses on a string: overlap time $= \dfrac{\text{gap}}{\text{closing speed}}$. Superposition: displacements add (Q5).
- Amplitude $= \dfrac{\text{peak-to-peak height}}{2}$. Speed of a crest $= \dfrac{\text{distance}}{\text{time}}$ (Q10, Q11).
- Number of wavelengths in a layer: $N = \dfrac{L}{\lambda_n} = \dfrac{L\,n}{\lambda}$ (Q13).
- Small angle on a screen at distance D: $y = D\tan\theta \approx D\sin\theta$ (Q14).
- Thin film, transmitted rays r_3 and r_4, with no net reflection phase shift: maxima $2L = m\dfrac{\lambda}{n_2}$, minima $2L = \left(m - \frac{1}{2}\right)\dfrac{\lambda}{n_2}$. With a net $\frac{1}{2}\lambda$ shift the two conditions swap (Q16). These are on the **current** F2026 sheet but were not on the F2023 sheet.
- Number of maxima around two sources: each m with $|m| < \frac{d}{\lambda}$ gives 2 points on a full circle. $|m| = \frac{d}{\lambda}$ gives 1 point on each side (on the axis) (Q15).
- Missing orders in a double-slit pattern: $\dfrac{d}{a} = m_{\text{missing}}$ (Q20).

Question 1 Adding waves of different amplitude (phasors)

1. These two waves travel along the same string:

$$y_1(x,t) = \left(\sqrt{2}\ \text{mm}\right)\sin(2\pi x - 200\pi t)$$

$$y_2(x,t) = (4\ \text{mm})\sin\left(2\pi x - 200\pi t + \frac{3\pi}{4}\ \text{rad}\right)$$

What is the amplitude of the resultant wave?

A) 3.16 mm
B) 9.0 mm
C) 3.0 mm
D) 10 mm

FROM THE FORMULA SHEET

$\tilde{A} = Ae^{i\theta} = A(\cos\theta + i\sin\theta)$

REMEMBER — not on the formula sheet

Phasor addition: add the real parts and the imaginary parts separately, then $|\tilde{A}| = \sqrt{\text{Re}^2 + \text{Im}^2}$.

Step 1. The waves have the same k and ω, so only the amplitudes and phases matter. Write them as phasors: $\sqrt{2}\angle 0$ and $4\angle\frac{3\pi}{4}$.

$$\tilde{A}_1 = \sqrt{2}(\cos 0 + i\sin 0) = 1.414 + 0i$$

$$\tilde{A}_2 = 4\left(\cos\tfrac{3\pi}{4} + i\sin\tfrac{3\pi}{4}\right) = 4(-0.7071) + 4(0.7071)i = -2.828 + 2.828i$$

Step 2. Add them: $\tilde{A} = (1.414 - 2.828) + 2.828i = -1.414 + 2.828i$

Step 3. Magnitude:

$$|\tilde{A}| = \sqrt{(-1.414)^2 + (2.828)^2} = \sqrt{2+8} = \sqrt{10} = 3.16\ \text{mm}$$

Final answer: **(A) 3.16 mm** ✓ Matches answer key

Question 2 Matching phases to snapshots

2. The figure 1 is a composite of three snapshots, each of a wave traveling along a particular string. The phases for the waves are given by (a) 2x − 4t, (b) 4x − 8t, and (c) 8x − 16t. Which phase corresponds to which wave in the figure?

A) (a) to 1; (b) to 2; (c) to 3
B) (a) to 2; (b) to 3; (c) to 1
C) (a) to 3; (b) to 2; (c) to 1
D) (a) to 3; (b) to 1; (c) to 2

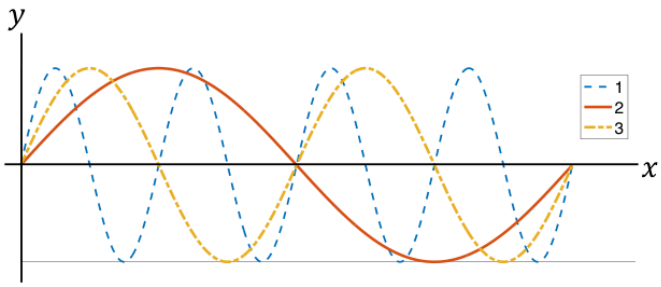


Figure 1

FROM THE FORMULA SHEET

Phase $= kx - \omega t, \quad k = \dfrac{2\pi}{\lambda}$

Step 1. Read k from each phase: (a) $k = 2$, (b) $k = 4$, (c) $k = 8$.

Step 2. Since $\lambda = \dfrac{2\pi}{k}$, the largest k has the shortest wavelength:

$$\lambda_a = \frac{2\pi}{2} = \pi, \quad \lambda_b = \frac{2\pi}{4} = \frac{\pi}{2}, \quad \lambda_c = \frac{2\pi}{8} = \frac{\pi}{4}$$

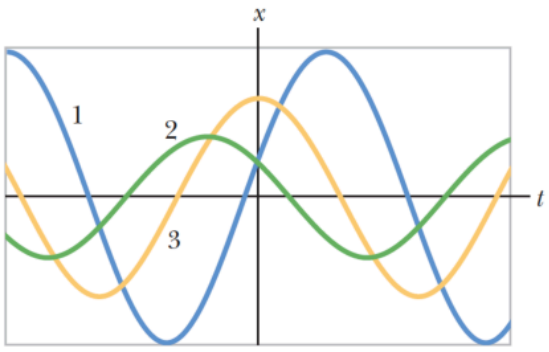
Step 3. In the figure, curve 2 has the longest wavelength, curve 3 has half of that, and curve 1 has a quarter. So (a) goes with 2, (b) with 3, and (c) with 1.

Final answer: **(B) (a)→2, (b)→3, (c)→1** ✓ Matches answer key

Question 3 Angular frequency from $x(t)$ curves

3. Figure 2 displays the x(t) curves for three experiments involving a specific spring-mass system oscillating in simple harmonic motion (SHM). Rank the curves according to the system's angular frequency and choose the correct answer.

A) $\omega_1 = \omega_2 = \omega_3$
B) $\omega_1 = \omega_2 < \omega_3$
C) $\omega_1 = \omega_2 > \omega_3$
D) $\omega_1 > \omega_3 > \omega_2$



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Figure 2

FROM THE FORMULA SHEET

$$\omega = \sqrt{\frac{k}{m}}, \quad \omega = \frac{2\pi}{T}$$

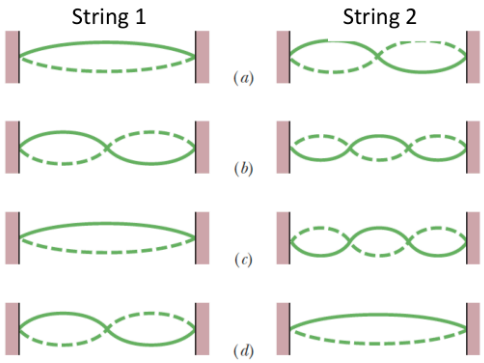
Step 1. It is the same spring–mass system in every experiment, so $\omega = \sqrt{\dfrac{k}{m}}$ is the same. The graph agrees: all three curves have the same period and differ only in amplitude and phase.

Final answer: **(A)** $\omega_1 = \omega_2 = \omega_3$ ✓ **Matches answer key**

Question 4 Same resonant frequency on two strings

4. Strings 1 and 2 have identical lengths and linear densities, but string 2 is under greater tension than string 1. Figure 3 shows four situation, (a) through (d), in which standing wave patterns exist on the two strings. In which situation is there the possibility that strings 1 and 2 are oscillating at the same resonant frequency?

A) (a)
B) (b)
C) (c)
D) (d)



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Figure 3

FROM THE FORMULA SHEET

$f = \dfrac{nv}{2L}$ (string fixed at both ends), $\quad v = \sqrt{\dfrac{\tau}{\mu}}$

Step 1. Same L and μ, but $\tau_2 > \tau_1$, so $v_2 > v_1$.

Step 2. For equal frequencies we need $\dfrac{n_1 v_1}{2L} = \dfrac{n_2 v_2}{2L}$, so $n_1 v_1 = n_2 v_2$. Since $v_2 > v_1$, string 2 must have **fewer** loops: $n_2 < n_1$.

Step 3. Only situation (d) has string 2 with fewer loops (1) than string 1 (2).

Final answer: **(D) (d)** ✓ **Matches answer key**

Question 5 Two pulses meeting

5. Two waves are traveling along a string in figure 4. The graph shows the position of the waves at time $t = 0.0$ s. One wave with a maximum amplitude of 0.5 cm is traveling toward the right at 0.5 cm/s. The second wave with a maximum amplitude of 2.0 cm is traveling toward the left at 2.0 cm/s. At what elapsed time will the two waves completely overlap and what will the maximum amplitude be at that time?

A) 2.0 s, 1.5 cm
B) 1.6 s, 2.5 cm
C) 1.0 s, 1.5 cm
D) 1.0 s, 2.5 cm

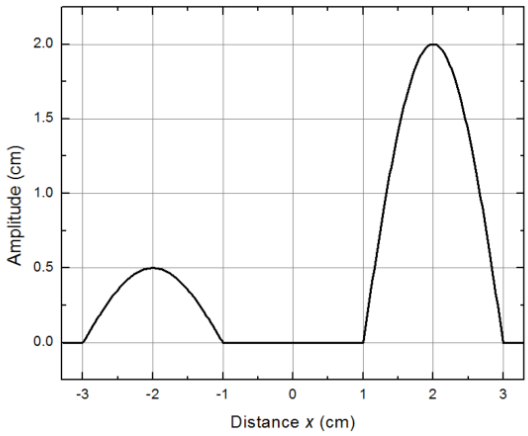


Figure 4

REMEMBER — not on the formula sheet
The gap between the two peaks closes at the sum of the speeds. When the pulses overlap, their displacements add (superposition).

Step 1. Peaks at $t = 0$: the small pulse is at $x = -2$ cm moving right at 0.5, and the big pulse is at $x = +2$ cm moving left at 2.0.

Step 2. Time for the peaks to meet:

$$t = \frac{\text{gap}}{\text{closing speed}} = \frac{2 - (-2)}{0.5 + 2.0} = \frac{4}{2.5} = 1.6 \text{ s}$$

Step 3. Maximum displacement when the peaks coincide: $0.5 + 2.0 = 2.5$ cm

Final answer: **(B) 1.6 s, 2.5 cm** ✓ Matches answer key

Question 6 What waves transport

6. A tsunami is a fast-moving ocean wave train that is produced during an earthquake. Consider such a wave initiated at center of the earthquake off the western coast of South America that reaches the Hawaiian Islands within 15 hours. Which one of the following statements concerning the tsunami is correct?

A) The tsunami carried water from the earthquake center to Hawaii, but it did not carry energy to Hawaii from South America.
B) The tsunami carried energy and water from the earthquake center to Hawaii.
C) The tsunami carried energy from the earthquake center to Hawaii, but it did not carry water to Hawaii from South America.
D) The tsunami did not carry energy or water from the earthquake center to Hawaii.

Step 1. A wave transports **energy** (and momentum). The medium only oscillates about its equilibrium position. The water near South America moves up and down and back and forth locally. It does not travel to Hawaii.

Final answer: **(C) Energy but not water** ✓ **Matches answer key**

Question 7 Doppler on a swing

7. A child is swinging back and forth with a constant period and amplitude (figure 5). Somewhere in front of the child, a stationary horn is emitting a constant tone of frequency f_s. Four points are labeled in the drawing to indicate positions along the arc as the child swings. At which position(s) will the child hear the lowest frequency for the sound from the whistle?

A) at **2** when moving toward **1**
B) at **2** when moving toward **3**
C) at **3** when moving toward **2**
D) at **3** when moving toward **4**

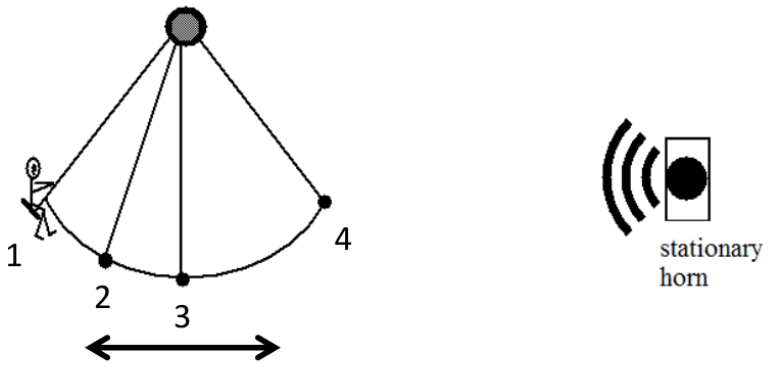


Figure 5

FROM THE FORMULA SHEET

$f' = f\,\dfrac{v \pm v_D}{v \pm v_S}$ (the detector is the child, and the source is at rest)

Step 1. With the source at rest, $f' = f\,\dfrac{v - v_D}{v}$ when moving **away**. This is lowest when the child moves away from the horn at the greatest speed.

Step 2. The horn is on the right. Moving toward point 2 means moving left, which is away from the horn. A swing is fastest at the bottom of the arc, which is point 3.

Step 3. So the lowest frequency is heard at point 3 while moving toward point 2.

Final answer: **(C) At 3 moving toward 2** ✓ **Matches answer key**

Question 8 Position from the acceleration graph

8. The acceleration $a(t)$ of a particle undergoing SHM is graphed in the Figure 6. The amplitude of the oscillation is $+x_m$. At point 5, particle is at:

A) $+x_m$
B) $-x_m$
C) Between $-x_m$ and 0
D) Between 0 and $+x_m$

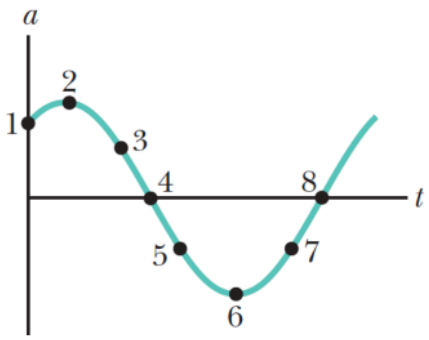


Figure 6

REMEMBER — not on the formula sheet

$a = -\omega^2 x$, so $x = -\dfrac{a}{\omega^2}$.

Step 1. At point 5 the graph shows $a < 0$, and its size is less than the maximum (point 6 is the most negative).

Step 2. So $x = -\dfrac{a}{\omega^2} > 0$, and since $|a| < a_{\max}$, $|x| < x_m$. The particle is between 0 and $+x_m$.

Final answer: **(D) Between 0 and $+x_m$** ✓ Matches answer key

Question 9 Ranking the first harmonics of pipes

9. Figure 7 shows three pipes, of lengths L and $2L$, in which the first harmonic has been set up. Rank them according to the frequency of the first harmonic, greatest first.

A) 1, 2, 3
B) 1, 3, 2
C) 2, 1, 3
D) 3, 2, 1

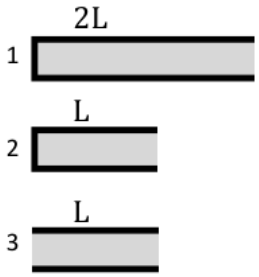


Figure 7

FROM THE FORMULA SHEET

Open both ends: $f = \dfrac{nv}{2L}$ Closed one end: $f = \dfrac{nv}{4L}$

Step 1. Pipe 1 has length $2L$ and is closed at one end: $f_1 = \dfrac{v}{4(2L)} = \dfrac{v}{8L}$

Step 2. Pipe 2 has length L and is closed at one end: $f_2 = \dfrac{v}{4L}$

Step 3. Pipe 3 has length L and is open at both ends: $f_3 = \dfrac{v}{2L}$

$$f_3 > f_2 > f_1$$

Final answer: **(D) 3, 2, 1** ✓ Matches answer key

Question 10 Amplitude from the figure

Questions 11, 12 and 13 are related to the following situation:

A sinusoidal wave moving along a string is shown twice in figure 8, as crest A travels in the positive direction of an x axis by distance $d = 6.0$ cm in 4.0 ms. The tick marks along the axis are separated by 10 cm; height $H = 6.00$ mm.

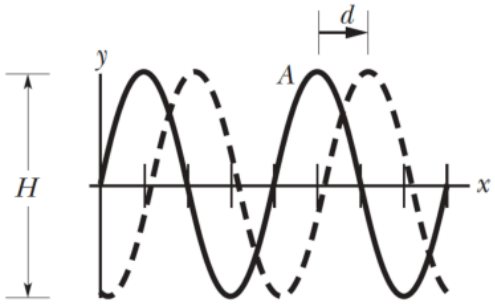


Figure 8

10. What is the amplitude of the wave?

A) 0.006 m
B) 0.060 m
C) 0.003 m
D) 0.006 m

REMEMBER — not on the formula sheet

Amplitude $= \dfrac{\text{crest-to-trough height}}{2}$

$$y_m = \frac{H}{2} = \frac{6.00 \text{ mm}}{2} = 3.00 \text{ mm} = 0.003 \text{ m}$$

Final answer: **(C) 0.003 m** ✓ Matches answer key

Question 11 Wave speed from crest motion

11. What is the speed of the wave?

A) 15 m/s
B) 1.5 m/s
C) 0.15 m/s
D) 150 m/s

REMEMBER — not on the formula sheet

Speed $= \dfrac{\text{distance}}{\text{time}}$

$$v = \frac{d}{\Delta t} = \frac{0.060\ \text{m}}{4.0 \times 10^{-3}\ \text{s}} = 15\ \tfrac{\text{m}}{\text{s}}$$

Final answer: **(A)** $15\ \frac{\text{m}}{\text{s}}$ ✓ Matches answer key

Question 12 Angular wave number

12. What is the angular wave number k?

A) 16 rad/m
B) 1.6 rad/m
C) 0.16 rad/m
D) 160 rad/m

FROM THE FORMULA SHEET

$k = \dfrac{2\pi}{\lambda}$

Step 1. Read λ from the figure. One full wavelength of the solid curve spans 4 tick marks, so $\lambda = 4 \times 10\ \text{cm} = 0.40\ \text{m}$.

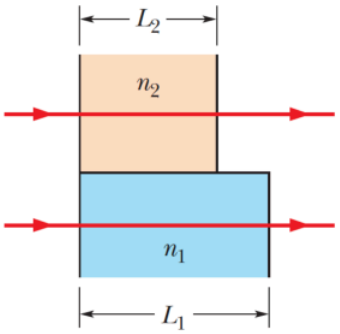
$$k = \frac{2\pi}{0.40} = 15.7 \approx 16\ \tfrac{\text{rad}}{\text{m}}$$

Final answer: **(A)** $16\ \frac{\text{rad}}{\text{m}}$ ✓ Matches answer key

Question 13 Phase difference through two plastic layers

13. Two waves of light in air, of wavelength λ = 800.0 nm, are initially in phase. They then both travel through a layer of plastic as shown in Figure 9, with L_1= 4.00 μm, L_2 = 3.50 μm, n_1 = 1.40, and n_2 = 1.60. What multiple of λ gives their phase difference after they both have emerged from the layers?

A) 0.833
B) 0.625
C) 0.714
D) 0.572



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Figure 9

FROM THE FORMULA SHEET

$$\lambda = \frac{\lambda_{vacuum}}{n}$$

REMEMBER — not on the formula sheet

Number of wavelengths that fit in a length L of material: $N = \dfrac{L}{\lambda_n} = \dfrac{Ln}{\lambda}$

Step 1. Bottom ray: it goes through $L_1 = 4.00\,\mu$m of n_1:

$$N_{bottom} = \frac{L_1 n_1}{\lambda} = \frac{(4.00)(1.40)}{0.800} = 7.000$$

Step 2. Top ray: it goes through $L_2 = 3.50\,\mu$m of n_2, then the remaining $L_1 - L_2 = 0.50\,\mu$m of air ($n = 1$):

$$N_{top} = \frac{(3.50)(1.60)}{0.800} + \frac{(0.50)(1.00)}{0.800} = 7.000 + 0.625 = 7.625$$

Step 3. Phase difference in wavelengths:

$$\Delta N = 7.625 - 7.000 = 0.625$$

Final answer: **(B) 0.625** ✓ Matches answer key

Question 14 Single-slit width from minima spacing

14. The distance between the first and sixth minima of a single- slit diffraction pattern is 0.35 mm with the screen 40 cm away from the slit, when light of wavelength 550 nm is used. What is the slit width?

A) 2.50 mm
B) 3.14 mm
C) 1.25 mm
D) 4.75 mm

FROM THE FORMULA SHEET

$a \sin\theta = m\lambda$ (minima, $m = 1, 2, 3, \ldots$)

REMEMBER — not on the formula sheet

Small angles: $y = D\tan\theta \approx D\sin\theta$, so the m-th minimum is at $y_m = \dfrac{m\lambda D}{a}$.

Step 1. Distance from the 1st to the 6th minimum:

$$y_6 - y_1 = \frac{(6-1)\lambda D}{a} = \frac{5\lambda D}{a}$$

Step 2. Solve for a:

$$a = \frac{5\lambda D}{y_6 - y_1} = \frac{5(550\times10^{-9})(0.40)}{0.35\times10^{-3}} = \frac{1.10\times10^{-6}}{3.5\times10^{-4}} = 3.14\times10^{-3}\ \text{m} = 3.14\ \text{mm}$$

Final answer: **(B) 3.14 mm** ✓ Matches answer key

Question 15 Counting maxima around two sources

15. In Figure 10, two radiofrequency point sources S_1 and S_2, separated by distance d = 4.0 m, are radiating in phase with λ = 0.50 m. A detector moves in a large circular path around the two sources in a plane containing them. How many maxima does it detect?

A) 16
B) 32
C) 8
D) 36

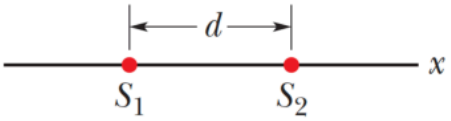


Figure 10

FROM THE FORMULA SHEET
$d \sin\theta = m\lambda$ (maxima)

Step 1. $\sin\theta = \dfrac{m\lambda}{d} = \dfrac{m(0.50)}{4.0} = \dfrac{m}{8}$. We need $|\sin\theta| \le 1$, so $m = -8, \ldots, 0, \ldots, +8$.

Step 2. Going around the full circle: each $|m| < 8$ gives a maximum in the upper half and another in the lower half, which is 2 points. $m = \pm 8$ gives $\sin\theta = \pm 1$, which lies on the x-axis, so each of those is only 1 point.

$$\underbrace{15}_{m=-7\ldots7} \times 2 + \underbrace{2}_{m=\pm8} = 32$$

Final answer: **(B) 32** ✓ Matches answer key

Question 16 Thin film, transmitted light minimum

16. In the Figure 11, light is incident perpendicularly on a thin layer of material 2 that lies between (thicker) materials 1 and 3. (The rays are tilted only for clarity.) Part of the light ends up in material 3 as ray r_3 (the light does not reflect inside material 2) and r_4 (the light reflects twice inside material 2). The waves of r_3 and r_4 interfere, and here we consider the type of interference to be either maximum (max) or minimum (min). For this situation, each following questions below refers to the indexes of refraction n_1, n_2, and n_3, the type of interference, the thin-layer thickness L in nanometers, and the wavelength λ in nanometers of the light as measured in air.
(The visible wavelengths range from 0.4μm to 0.7μm)

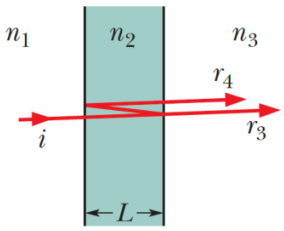


Figure 11

What is the wavelength in the visible range for the values in table below?

n_1	n_2	n_3	Type	L	λ
1.32	1.75	1.39	Min	325	

A) 455 nm
B) 423 nm
C) 478 nm
D) 497 nm

FROM THE FORMULA SHEET

$\lambda = \dfrac{\lambda_{vacuum}}{n}$. Reflection off a lower index gives 0 shift. Reflection off a higher index gives 0.5 wavelength.

REMEMBER — not on the formula sheet

Transmission thin-film conditions: path difference $2L$. With no net reflection shift, a **minimum** needs $2L = \left(m - \frac{1}{2}\right)\dfrac{\lambda}{n_2}$, $m = 1, 2, 3, \ldots$. (This is on the current F2026 sheet.)

Step 1. Phase shifts for r_4: it reflects at the 2|3 interface ($1.75 \to 1.39$, off a lower index, so 0) and at the 2|1 interface ($1.75 \to 1.32$, off a lower index, so 0). r_3 does not reflect. So the net reflection shift is 0, and only the extra path $2L$ matters.

Step 2. Minimum condition:

$$2L = \left(m - \tfrac{1}{2}\right)\frac{\lambda}{n_2} \;\Rightarrow\; \lambda = \frac{2Ln_2}{m - \frac{1}{2}} = \frac{2(325)(1.75)}{m - \frac{1}{2}} = \frac{1137.5}{m - \frac{1}{2}}\ \text{nm}$$

Step 3. Try values of m: $m = 1$ gives 2275 nm, $m = 2$ gives 758 nm, $m = 3$ gives $\dfrac{1137.5}{2.5} = 455$ nm (visible ✓), and $m = 4$ gives 325 nm (too short).

Final answer: **(A) 455 nm** ✓ Matches answer key

Question 17 Joined strings, transmitted power

17. A string having linear mass density μ_1 is joined to a second, lighter string with mass density μ_2 and placed under tension τ. A sinusoidal wave carrying power P is launched into string 1, resulting in a wave in string 2 carrying power $\frac{9}{25}P$. What is μ_2/μ_1?

A) $\frac{3}{5}$
B) $\frac{5}{3}$
C) 8
D) $\frac{1}{8}$

FROM THE FORMULA SHEET

$$Z = \sqrt{\mu\tau}, \quad R = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad T = \frac{2Z_1}{Z_1 + Z_2}, \quad R^2 + \frac{Z_2}{Z_1}T^2 = 1$$

Step 1. The term $\frac{Z_2}{Z_1}T^2$ in the energy equation is the fraction of power transmitted. Set it equal to $\frac{9}{25}$. Easier: use the reflected part, $R^2 = 1 - \frac{9}{25} = \frac{16}{25}$, so $|R| = \frac{4}{5}$.

Step 2. String 2 is lighter, so $Z_2 < Z_1$ and $R > 0$:

$$\frac{Z_1 - Z_2}{Z_1 + Z_2} = \frac{4}{5} \;\Rightarrow\; 5Z_1 - 5Z_2 = 4Z_1 + 4Z_2 \;\Rightarrow\; Z_1 = 9Z_2 \;\Rightarrow\; \frac{Z_2}{Z_1} = \frac{1}{9}$$

Step 3. Check: $T = \frac{2}{1 + \frac{1}{9}} = \frac{9}{5}$, so $\frac{Z_2}{Z_1}T^2 = \frac{1}{9} \cdot \frac{81}{25} = \frac{9}{25}$ ✓

Step 4. Same tension, so $\frac{Z_2}{Z_1} = \sqrt{\frac{\mu_2}{\mu_1}}$:

$$\frac{\mu_2}{\mu_1} = \left(\frac{1}{9}\right)^2 = \frac{1}{81}$$

NOTE ON THE ANSWER KEY

The key highlights option D, which is printed as $\frac{1}{8}$. The physics gives $\frac{1}{81}$. D is the only option for a much lighter string, so the "1/8" is almost certainly a typo for 1/81. Pick D, but know that the true value is $\frac{1}{81}$.

Final answer: **(D)** $\frac{\mu_2}{\mu_1} = \frac{1}{81}$ **(printed as $\frac{1}{8}$)** **✓ Key = D (typo noted)**

Question 18 Change in sound level

18. If you were listening to an intense conversation at 80 dB, and you move to a location where the sound intensity decreases by a factor of 100, what would be the new sound intensity level?

A) 70 dB
B) 60 dB
C) 50 dB
D) 40 dB

FROM THE FORMULA SHEET

$$\Delta\beta = (10\text{ dB})\log_{10}\left(\frac{I_2}{I_1}\right)$$

$$\Delta\beta = 10\log_{10}\left(\frac{1}{100}\right) = 10(-2) = -20\text{ dB} \;\Rightarrow\; \beta_2 = 80 - 20 = 60\text{ dB}$$

Final answer: **(B) 60 dB** ✓ Matches answer key

Question 19 Fundamental of an open–open tube

19. A tube, which remains open at both ends, sustains a standing wave with harmonics occurring at 200 Hz and 300 Hz, and no harmonics in between. What is the fundamental frequency of this tube?

A) 250 Hz
B) 200 Hz
C) 100 Hz
D) 50 Hz

FROM THE FORMULA SHEET

$$f = \frac{nv}{2L} = nf_1,\ n = 1, 2, 3, \ldots$$

Step 1. Every integer n is allowed, so consecutive harmonics differ by exactly f_1:

$$f_1 = 300 - 200 = 100\text{ Hz}$$

Step 2. Check: $200 = 2f_1$ and $300 = 3f_1$, which are consecutive harmonics ✓

Final answer: **(C) 100 Hz** ✓ Matches answer key

Question 20 Slit width from missing orders

20. The figure 12 below shows the intensity of 500 nm light projected through two narrow slits onto a distant screen. The labels on the x axis indicate the fringe order of the 2-slit pattern. If the two slits are 2.0 mm apart, what is the width of each slit?

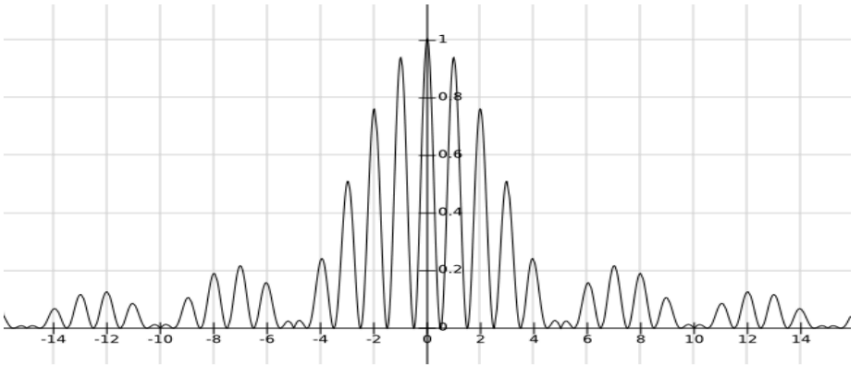


Figure 12

A) 1.0 mm
B) 2.0 mm
C) 2.5 mm
D) 0.4 mm

FROM THE FORMULA SHEET

$d \sin\theta = m\lambda$ (double-slit maxima), $a \sin\theta = m\lambda$ (single-slit minima)

REMEMBER — not on the formula sheet

A double-slit maximum is missing where it lands on a single-slit minimum. Dividing the two equations gives $\frac{d}{a} = m_{\text{missing}}$ for the first missing order.

Step 1. In the figure, the fringes at orders ± 5 (and ± 10) are missing, so the first single-slit minimum is at the 5th double-slit order.

Step 2. Set the angles equal:

$$\frac{d \sin\theta}{a \sin\theta} = \frac{5\lambda}{1\lambda} \Rightarrow a = \frac{d}{5} = \frac{2.0 \text{ mm}}{5} = 0.40 \text{ mm}$$

Final answer: **(D) 0.4 mm** ✓ Matches answer key